

# **Predicting Used Car Prices With Machine Learning**

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# Introduction

This project focuses on predicting the price of used cars using machine learning techniques. The goal is to develop models that can accurately estimate a vehicle's value based on features such as mileage, engine size, fuel type, and other attributes. Accurately predicting car prices is important for both buyers and sellers, as it helps ensure fair market value and supports informed decision-making. One challenge in this problem is that multiple factors influence price, and their relationships may not always be straightforward or purely linear.

To address this, three machine learning models were implemented: Linear Regression, Ridge Regression, and Random Forest. Linear Regression provides a simple baseline model, Ridge Regression extends it with regularization to reduce potential overfitting, and Random Forest introduces a more complex, non-linear approach. By comparing these models, this project aims to evaluate how different learning methods perform on real-world data and determine which approach is most effective for predicting car prices.

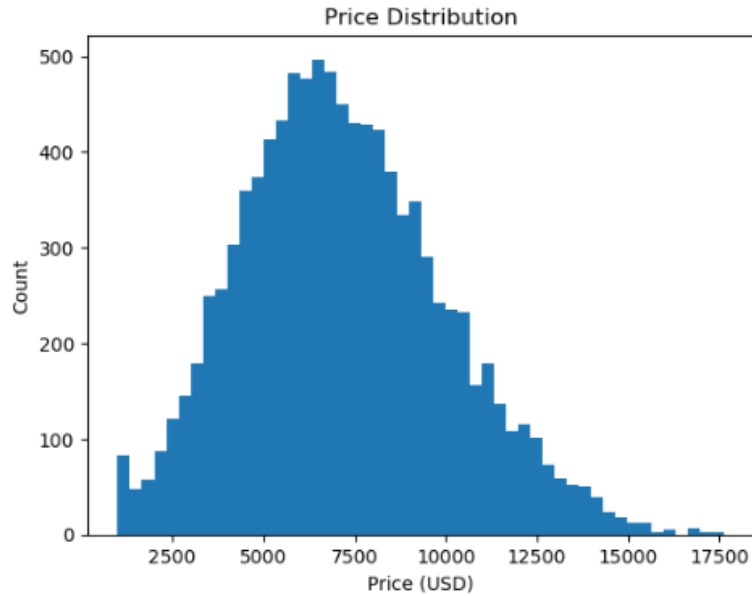
## Data

### 1. Introducing the Data

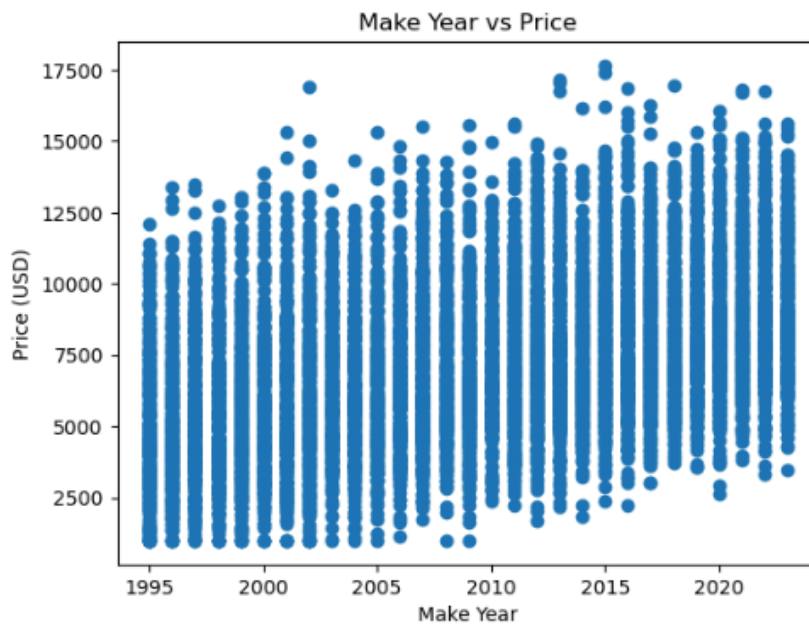
The dataset used in this project contains 10,000 entries with 12 features describing various attributes of used cars. These features include numerical variables such as mileage (km per liter), engine size (cc), number of previous owners, and reported accidents, as well as categorical variables such as fuel type, brand, transmission, color, service history, and insurance status. The target variable is the price of the car, measured in USD, which is the value the models aim to predict. The dataset covers a wide range of vehicles, with prices ranging from approximately \$1,000 to over \$17,000, providing sufficient variation for meaningful machine learning analysis. Initial inspection of the dataset showed that most columns contain complete data, with only minor missing values in the service history feature. Overall, the dataset is well-structured and suitable for regression modeling.

## 2. Visual Analysis

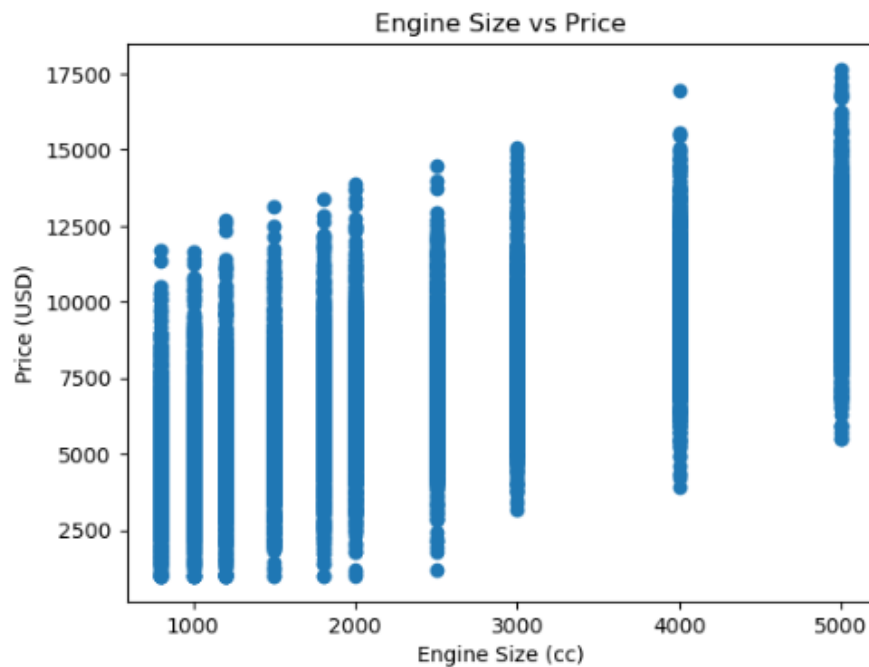
The price distribution shows a roughly normal shape with a slight right skew, indicating that most vehicles fall within a mid-range price while fewer vehicles have very high values.



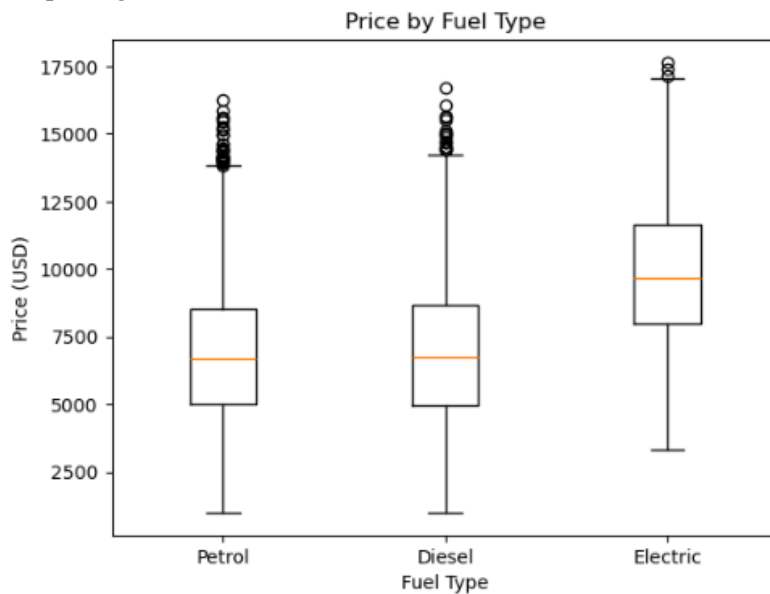
The relationship between make year and price reveals a clear positive trend, where newer vehicles tend to have higher prices. This suggests that vehicle age plays a significant role in determining value.



Engine size also shows a strong positive relationship with price. Vehicles with larger engines are generally more expensive, although the clustering of points indicates that engine size alone does not fully determine price.



Additionally, a comparison of prices across fuel types shows that electric vehicles tend to have higher median prices than petrol and diesel vehicles. This highlights how categorical features can significantly influence pricing.



### **3. Data Preprocessing**

Before training the machine learning models, the dataset required preprocessing to convert categorical variables into a numerical format. This was done using one-hot encoding, which transforms categorical features such as fuel type, brand, transmission, and service history into binary indicator variables. Numerical features were standardized to ensure that all variables are on a similar scale, preventing features with larger magnitudes from dominating the model. The dataset was then split into training, validation, and testing sets using a 60/20/20 split. The training set was used to fit the models, the validation set was used for model comparison and tuning, and the test set was reserved for final evaluation. This preprocessing pipeline ensures that the models can effectively learn from both numerical and categorical data while maintaining generalization to unseen data.

## **Machine Learning Methods**

In this project, three machine learning models were implemented to predict the price of used cars: Linear Regression, Ridge Regression, and Random Forest. These models were selected to compare simple linear approaches with more complex non-linear methods and evaluate their effectiveness on the dataset.

### **1. Linear Regression**

Linear Regression is a supervised learning algorithm used to model the relationship between input features and a continuous target variable. The model assumes that the target variable can be expressed as a linear combination of the input features, where each feature is assigned a coefficient representing its contribution to the prediction. The model is trained by minimizing the sum of squared differences between predicted and actual values, commonly referred to as the least squares method. This optimization process ensures that the fitted line best represents the relationship between variables by reducing overall prediction error (Pedregosa et al.).

Linear Regression was chosen as a baseline model because of its simplicity, interpretability, and effectiveness when relationships between variables are approximately linear. Given that features such as engine size and vehicle age are expected to have direct effects on price, this model provides a strong foundation for comparison.

### **2. Ridge Regression**

Ridge Regression is an extension of Linear Regression that incorporates L2 regularization. In addition to minimizing the sum of squared errors, Ridge Regression adds a penalty term proportional to the square of the magnitude of the coefficients. This penalty discourages large coefficient values and helps prevent overfitting (Pedregosa et al.). By constraining the model parameters, Ridge Regression

reduces the impact of multicollinearity, which occurs when input features are highly correlated. This results in more stable and generalized predictions when compared to standard Linear Regression.

Ridge Regression was included in this project to evaluate whether regularization improves model performance. By comparing it with Linear Regression, it is possible to determine whether overfitting or feature correlation significantly affects the dataset.

### **3. Random Forest**

Random Forest is an ensemble learning method that constructs multiple decision trees during training and combines their predictions through averaging. Each tree is trained on a random subset of the data and considers a random subset of features when making splits. This randomness reduces variance and improves model robustness (Pedregosa et al.). Unlike linear models, Random Forest does not assume a specific functional relationship between input features and the target variable. Instead, it can capture complex, non-linear interactions by recursively partitioning the data into smaller subsets based on feature values.

Random Forest was selected to determine whether a more complex, non-linear model could outperform linear approaches. Its ability to model interactions between variables makes it particularly useful for datasets where relationships are not strictly linear.

To evaluate the performance of each model, three metrics were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  score. MAE measures the average absolute difference between predicted and actual values, providing a straightforward interpretation of prediction error. RMSE penalizes larger errors more heavily and is useful for identifying models that produce large deviations. The  $R^2$  score measures the proportion of variance in the target variable that is explained by the model, indicating overall goodness of fit. These metrics provide a comprehensive evaluation of model performance and allow for meaningful comparison between different approaches.

## **Results**

To evaluate the performance of the models, the dataset was divided into training, validation, and testing sets using a 60/20/20 split. The training set was used to fit each model, while the validation set was

used to compare model performance during development. The test set was reserved for final evaluation to ensure that results reflect performance on unseen data.

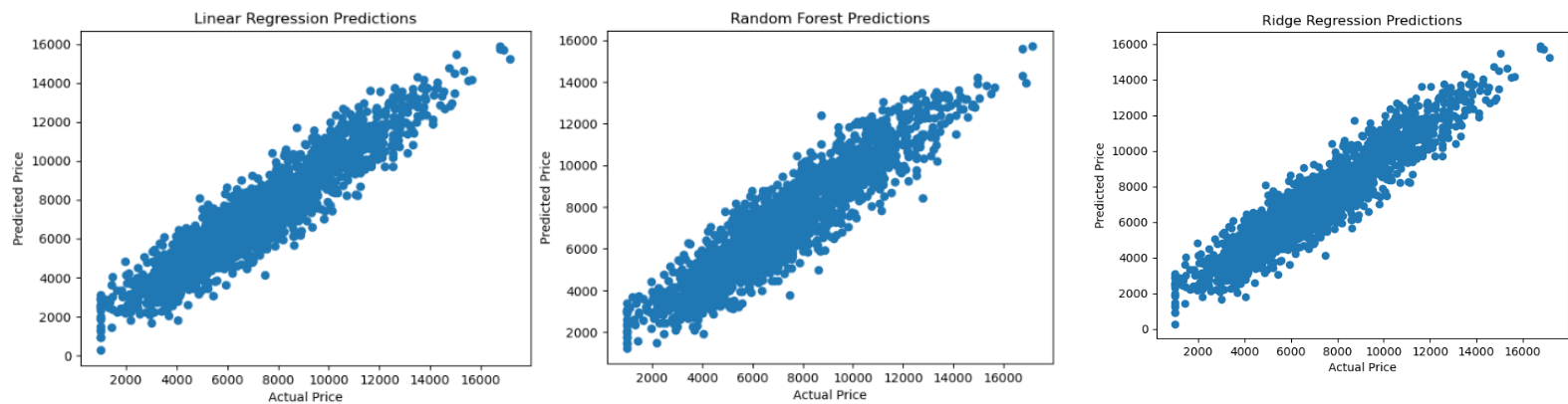
All models were trained using the same preprocessing pipeline, which included one-hot encoding for categorical variables and standardization for numerical features. This ensured a fair comparison between models by maintaining consistent data preparation.

## Model Results

The performance of each model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  score on the test dataset.

|   | Model             | Test MAE   | Test RMSE   | Test R2  |
|---|-------------------|------------|-------------|----------|
| 0 | Linear Regression | 790.969730 | 990.650945  | 0.876640 |
| 1 | Ridge Regression  | 790.969019 | 990.665868  | 0.876637 |
| 2 | Random Forest     | 876.609133 | 1091.281940 | 0.850306 |

Linear Regression and Ridge Regression achieved nearly identical results, both outperforming the Random Forest model across all evaluation metrics. Random Forest showed higher error values and a lower  $R^2$  score, indicating reduced predictive accuracy.



## Observation and Analysis

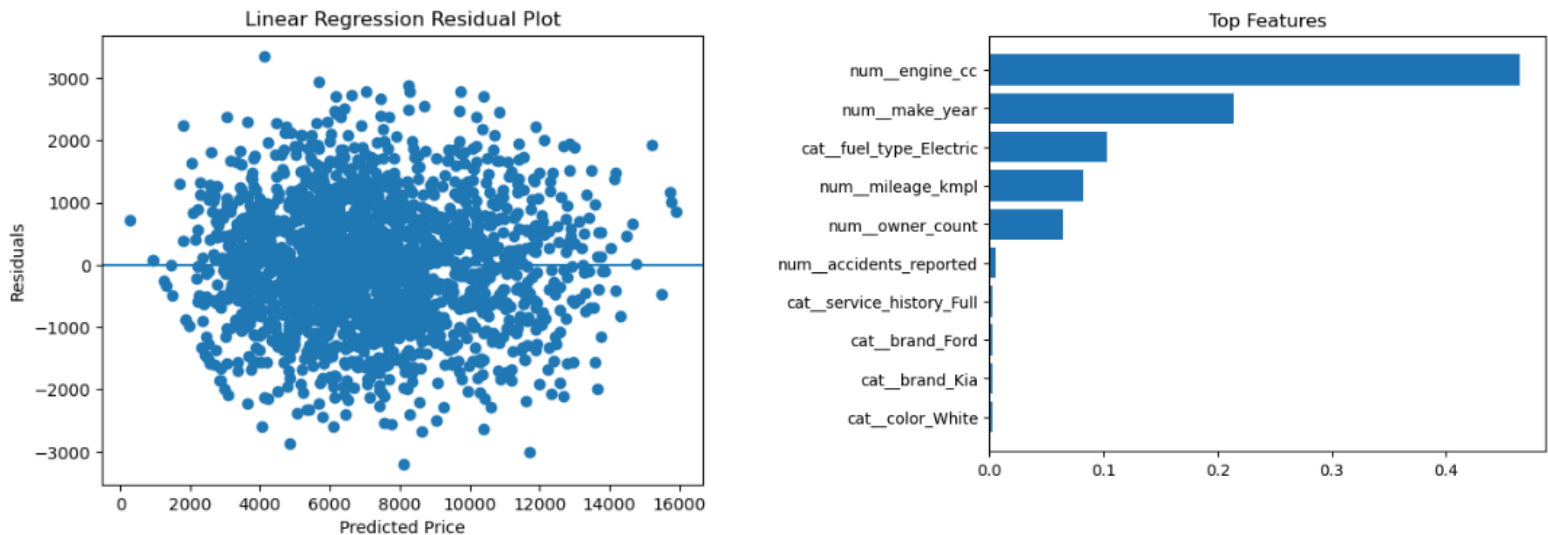
The results indicate that Linear Regression and Ridge Regression provide the most accurate predictions for this dataset. This suggests that the relationships between the input features and the target variable are largely linear in nature. The similarity in performance between Linear Regression and Ridge Regression indicates that regularization had minimal effect, suggesting that overfitting and multicollinearity are not significant issues in the dataset.

The Random Forest model, while capable of capturing complex non-linear relationships, performed worse in this case. This suggests that the additional complexity of the model is not necessary for this dataset and may introduce unnecessary variance without improving accuracy.

The prediction plots for all three models show that Linear Regression and Ridge Regression produce predictions that closely follow the ideal diagonal trend, indicating strong agreement between predicted and actual values. In contrast, the Random Forest predictions show slightly greater dispersion, particularly at higher price ranges, indicating less consistent accuracy.

The residual plot for Linear Regression shows that errors are distributed relatively evenly around zero, supporting the assumption that the model is appropriate for this dataset.

Feature importance analysis from the Random Forest model reveals that engine size and make year are the most influential features in predicting car price. This aligns with the earlier visual analysis and confirms that these features play a key role in determining vehicle value.



## Experiments Supporting Analysis

To further validate the results, several additional experiments were conducted. First, I compared model performance after removing key features such as engine size and make year. This resulted in a noticeable increase in prediction error, confirming that these features play a critical role in determining vehicle price.

Second, I analyzed model performance on higher-priced vehicles to examine how each model handles extreme values. The results showed that Random Forest exhibited greater error variability, while Linear Regression maintained more consistent predictions.

Additionally, I compared actual vs predicted plots for each model. Linear and Ridge Regression predictions closely followed the ideal diagonal trend, while Random Forest showed greater dispersion, especially at higher price ranges. These experiments reinforce the conclusion that simpler linear models are better suited for this dataset, as the relationships between features and price are largely linear.

Original RMSE: 990.650945499776  
Reduced RMSE: 2498.883161014641

Linear RMSE (high price): 1166.4826942938143  
Random Forest RMSE (high price): 1472.7658805134274

## Conclusion

In this project, machine learning techniques were applied to predict the price of used cars using a dataset containing 10,000 entries and multiple vehicle attributes. Three models were implemented: Linear Regression, Ridge Regression, and Random Forest. The results showed that Linear Regression and Ridge Regression consistently outperformed Random Forest, achieving lower error values and higher  $R^2$  scores. This indicates that the relationships between the features and the target variable are largely linear, making simpler models more effective for this problem. Through additional experiments, it was confirmed that key features such as engine size and make year play a significant role in determining car price, as removing these features led to a noticeable decrease in model performance. Furthermore, the similarity between validation and test results demonstrated that the models generalize well and do not suffer from significant overfitting. One of the main lessons learned from this project is that more complex models do not always lead to better performance. While Random Forest is capable of capturing non-linear relationships, it did not outperform linear models in this case. This highlights the importance of understanding the structure of the data when selecting machine learning models. A challenge encountered during this project was determining which features and models would provide the best predictive performance. Through experimentation and analysis, it became clear that simpler approaches were more appropriate for the dataset.

## References

“Used Car Price Prediction Dataset.” *Kaggle*,  
<https://www.kaggle.com/datasets/therohithanand/used-car-price-prediction>. Accessed 1 May 2026.

Pedregosa, F., et al. “Scikit-learn: Machine Learning in Python.” *Journal of Machine Learning Research*, vol. 12, 2011, pp. 2825–2830.

*Scikit-learn Documentation*. *scikit-learn*, <https://scikit-learn.org/stable/>. Accessed 1 May 2026.

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